Chapter 8 Homework: Heteroskedasticity

Claire

2026-05-29

```
library(lmtest)
library(sandwich)
```

Exercise 8.16

A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation:

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/vacation.rdata"))
head(vacation)
```

	miles <int>	income <int>	age <int>	kids <int>
1	902	41	26	0
2	491	31	38	3
3	1841	87	40	2
4	406	54	48	4
5	0	77	43	4
6	1899	70	55	2

6 rows

```
str(vacation)
```

```
## 'data.frame':	200 obs. of  4 variables:
##  $ miles : int  902 491 1841 406 0 1899 724 1404 1367 560 ...
##  $ income: int  41 31 87 54 77 70 43 40 69 38 ...
##  $ age   : int  26 38 40 48 43 55 24 39 55 29 ...
##  $ kids  : int  0 3 2 4 4 2 0 0 2 2 ...
##  - attr(*, "var.labels")= chr [1:4] "miles traveled per year" "annual income ($100
0)" "average age of adult members of household" "number of children in household"
```

Part (a): OLS estimation and 95% CI for KIDS

effect

```
model_vac <- lm(miles ~ income + age + kids, data = vacation)
summary(model_vac)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1198.14  -295.31    17.98   287.54  1549.41
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -391.548    169.775  -2.306   0.0221 *
## income        14.201      1.800   7.889 2.10e-13 ***
## age           15.741      3.757   4.189 4.23e-05 ***
## kids         -81.826     27.130  -3.016   0.0029 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 452.3 on 196 degrees of freedom
## Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
## F-statistic: 33.75 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
ci <- confint(model_vac, level = 0.95)
cat("\n95% CI for KIDS coefficient:\n")
```

```
##
## 95% CI for KIDS coefficient:
```

```
print(ci["kids", ])
```

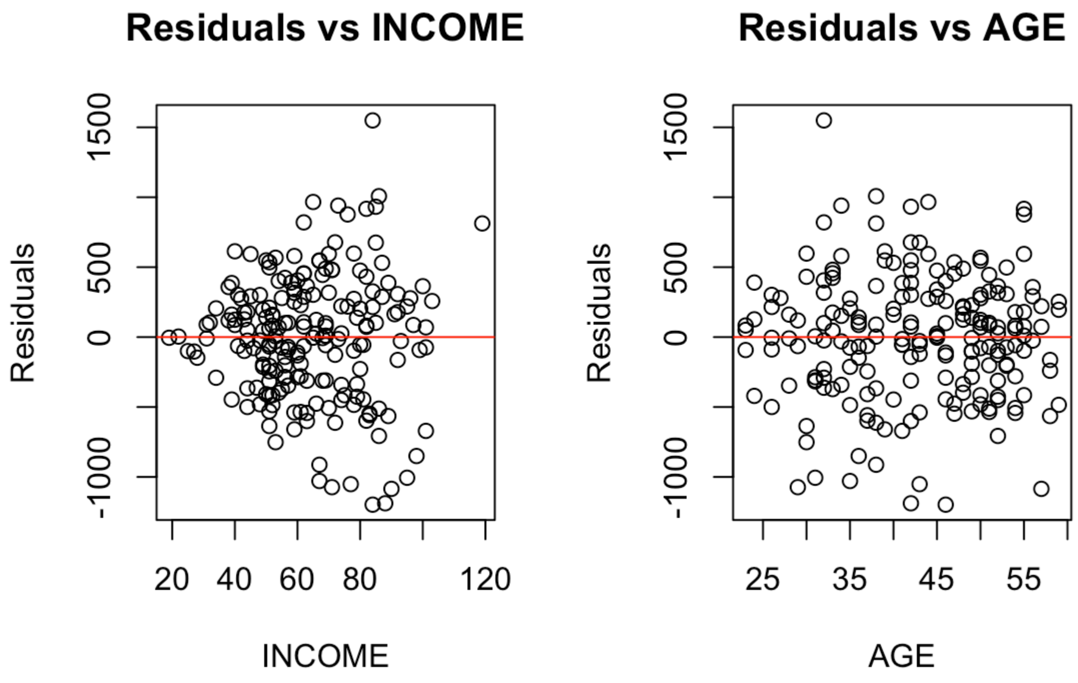
```
##      2.5 %     97.5 %
## -135.32981  -28.32302
```

Interpretation: Holding INCOME and AGE constant, each additional child is associated with a change of -81.83 miles driven per year.

Part (b): Residual plots vs INCOME and AGE

```
par(mfrow = c(1, 2))
plot(vacation$income, residuals(model_vac),
     xlab = "INCOME", ylab = "Residuals", main = "Residuals vs INCOME")
abline(h = 0, col = "red")

plot(vacation$age, residuals(model_vac),
     xlab = "AGE", ylab = "Residuals", main = "Residuals vs AGE")
abline(h = 0, col = "red")
```




```
par(mfrow = c(1, 1))
```

Observation: If the spread of residuals increases with INCOME, this indicates heteroskedasticity.

Part (c): Goldfeld–Quandt test (sorted by INCOME)

```
# Sort by income
vac_sorted <- vacation[order(vacation$income), ]
N    <- nrow(vac_sorted)
n90  <- 90

# Lower 90 observations
lower90 <- vac_sorted[1:n90, ]
upper90 <- vac_sorted[(N - n90 + 1):N, ]

model_lower <- lm(miles ~ income + age + kids, data = lower90)
model_upper <- lm(miles ~ income + age + kids, data = upper90)

SSE_lower <- sum(residuals(model_lower)^2)
SSE_upper <- sum(residuals(model_upper)^2)

df1 <- n90 - 4; df2 <- n90 - 4
GQ  <- (SSE_upper / df1) / (SSE_lower / df2)

p_val <- pf(GQ, df1 = df1, df2 = df2, lower.tail = FALSE)

cat("Goldfeld-Quandt F statistic:", round(GQ, 4), "\n")
```

```
## Goldfeld-Quandt F statistic: 3.1041
```

```
cat("p-value:", round(p_val, 4), "\n")
```

```
## p-value: 0
```

```
cat("5% critical value:", round(qf(0.95, df1, df2), 4), "\n")
```

```
## 5% critical value: 1.4286
```

```
cat("Reject H0?", GQ > qf(0.95, df1, df2), "\n")
```

```
## Reject H0? TRUE
```

Part (d): OLS with heteroskedasticity-robust standard errors

```
robust_se <- coeftest(model_vac, vcov = vcovHC(model_vac, type = "HC1"))
print(robust_se)
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) -391.5480   142.6548 -2.7447 0.0066190 **
## income        14.2013     1.9389  7.3246 6.083e-12 ***
## age           15.7409     3.9657  3.9692 0.0001011 ***
## kids         -81.8264    29.1544 -2.8067 0.0055112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ci_robust <- coefci(model_vac, vcov = vcovHC(model_vac, type = "HC1"))
cat("\n95% CI for KIDS (robust):\n")
```

```
##
## 95% CI for KIDS (robust):
```

```
print(ci_robust["kids", ])
```

```
##      2.5 %     97.5 %
## -139.32297  -24.32986
```

Part (e): GLS assuming $\sigma_i^2 = \sigma^2 \cdot INCOME_i^2$

Under this assumption, divide all variables by $INCOME_i$:

```
# GLS with variance proportional to INCOME^2
model_gls <- lm(miles ~ income + age + kids, data = vacation,
                weights = 1 / vacation$income^2)
summary(model_gls)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/vacation$in
come^2)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1907  -4.9555   0.2488   4.3832  18.5462
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -424.996    121.444  -3.500 0.000577 ***
## income        13.947      1.481   9.420  < 2e-16 ***
## age           16.717      3.025   5.527 1.03e-07 ***
## kids         -76.806     21.848  -3.515 0.000545 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.765 on 196 degrees of freedom
## Multiple R-squared:  0.4573, Adjusted R-squared:  0.449
## F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
# CI for KIDS (conventional GLS SE)
ci_gls <- confint(model_gls, level = 0.95)
cat("\n95% CI for KIDS (GLS, conventional):\n")
```

```
##
## 95% CI for KIDS (GLS, conventional):
```

```
print(ci_gls["kids", ])
```

```
##      2.5 %     97.5 %
## -119.89450  -33.71808
```

```
# CI for KIDS (robust GLS SE)
ci_gls_rob <- coefci(model_gls, vcov = vcovHC(model_gls, type = "HC1"))
cat("\n95% CI for KIDS (GLS, robust):\n")
```

```
##
## 95% CI for KIDS (GLS, robust):
```

```
print(ci_gls_rob["kids", ])
```

```
##      2.5 %     97.5 %
## -121.41339  -32.19919
```

```
# Comparison of all interval estimates
cat("\n--- Summary Comparison for KIDS Coefficient ---\n")
```

```
##
## --- Summary Comparison for KIDS Coefficient ---
```

```
cat("OLS:          [", round(ci["kids",1],2), ",", round(ci["kids",2],2), "]\n")
```

```
## OLS:          [ -135.33 , -28.32 ]
```

```
cat("OLS Robust:   [", round(ci_robust["kids",1],2), ",", round(ci_robust["kids",2],
2), "]\n")
```

```
## OLS Robust:   [ -139.32 , -24.33 ]
```

```
cat("GLS:          [", round(ci_gls["kids",1],2), ",", round(ci_gls["kids",2],2), "]
\n")
```

```
## GLS:          [ -119.89 , -33.72 ]
```

```
cat("GLS Robust:   [", round(ci_gls_rob["kids",1],2), ",", round(ci_gls_rob["kids",
2],2), "]\n")
```

```
## GLS Robust:   [ -121.41 , -32.2 ]
```

Interpretation: The GLS estimator assumes $\sigma_i^2 = \sigma^2 INCOME_i^2$. If this form is correct, GLS is more efficient than OLS, yielding narrower confidence intervals.